NUCLEAR PROGRAMME

A CRITICAL LEAP

On April 6, India's nuclear energy programme officially stepped into its second stage with the Prototype Fast Breeder Reactor at Kalpakkam achieving first criticality—the point at which a sustained and controlled fission chain reaction begins. Once it is fully operational, India will become only the second country after Russia to run a commercial fast breeder reactor. A closer look at the science and strategy behind this milestone

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PROTOTYPE FAST BREEDER REACTOR

Designed by: Indira Gandhi Centre for Atomic Research

Commissioned by: Bharatiya Nabhikiya Vidyut Nigam Ltd (Department of Atomic Energy)

Location: Kalpakkam Nuclear Complex, Tamil Nadu

500 MWe
i.e. Megawatt electrical capacity

₹8,180 CRORE
Project cost (original estimate: ₹3,492 crore)

2004: Prototype's construction begins

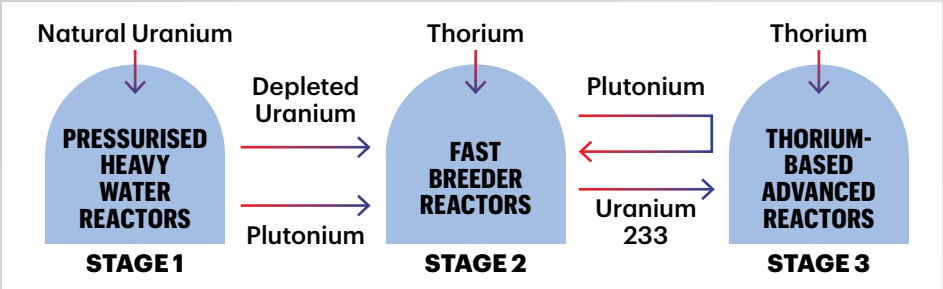
2024: Fuel loading completed

Apr. 6, 2026: First criticality achieved

Dec. 2026 (expected): Full power generation

INDIA'S THREE-STAGE NUCLEAR PROGRAMME

Fast breeder reactors are a key element of India's long-term nuclear strategy. With limited uranium and abundant thorium, the plan relies on recycling nuclear fuel



STAGE 1

Pressurised heavy water reactors (PHWRs)

Input: Natural uranium—0.7% fissile U-235 (sustains a chain reaction) and 99.3% fertile U-238 (can be converted into fissile fuel in a reactor)

Output: U-235 undergoes fission to produce electricity; some U-238 absorbs neutrons, converts into fissile Plutonium-239

STAGE 2

Fast breeder reactors

Input: Mixed Oxide, or MOX fuel (blend of Pu-239 bred in stage 1 and depleted uranium) + U-238 blanket (layer) around the core

Output: Fast neutrons turn U-238 into Pu-239, so the reactor produces more fuel than it uses while generating power. Eventually, Thorium-232 blanket will be used to breed U-233 for Stage 3

STAGE 3

Thorium-based reactors

Input: Fissile U-233 (bred in Stage 2) + fertile Th-232 (occurs in monazite, a phosphate mineral abundant in India's beach sands)

Output: U-233 undergoes fission to produce energy; Th-232 converts into U-233, moving toward a near self-sustaining fuel cycle

How a nuclear reactor generates electricity

